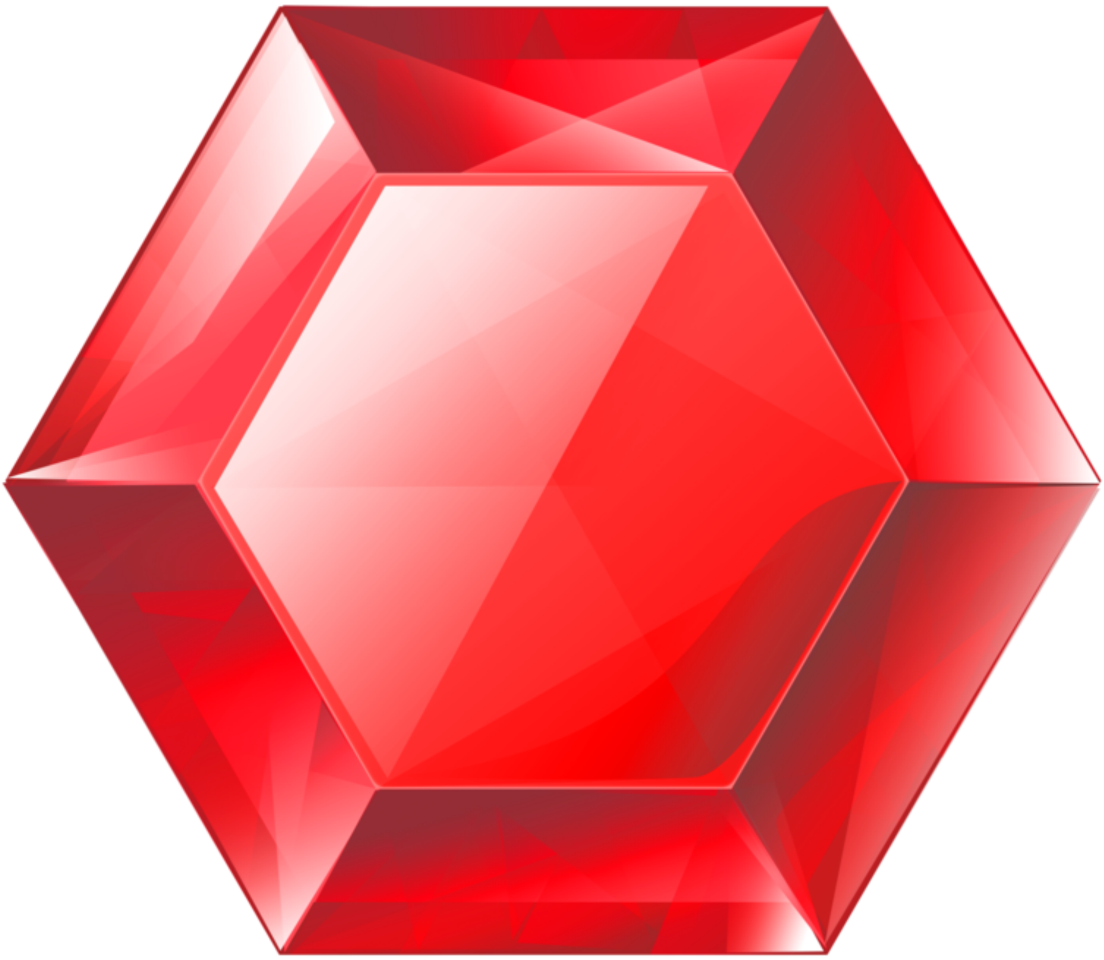
gemstone-rs Cookbook

Task-focused Rust recipes for GemStone/S



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Cookbook

This cookbook is a collection of direct `gemstone-rs` recipes. Each recipe is small enough to copy into a Rust CLI, service, or maintenance tool.

Recipe 1: Open a Session and Evaluate Smalltalk

```
use gemstone_rs::{Config, Session};

let mut session = Session::login(Config::from_env()?)?;
println!("{:?}", session.eval("3 factorial")?);
```

Use this when you need the smallest live sanity check.

Recipe 2: Verify `3 + 4`

```
use gemstone_rs::{Config, Session, Value};

let mut session = Session::login(Config::from_env()?)?;
assert_eq!(session.eval("3 + 4")?, Value::SmallInt(7));
```

This is the test to keep in every live smoke lane.

Recipe 3: Write and Read `UserGlobals`

```
use gemstone_rs::{Config, Oop, Session};

let mut session = Session::login(Config::from_env()?)?;
let key = "GemStoneRsCookbook";
let value = session.new_string("stored from Rust")?;
session.global_put(key, value)?;

let stored = session.global_get(key)?;
println!("{}", session.fetch_string(stored)?);

session.global_put(key, Oop::NIL)?;
```

Recipe 4: Commit a Write Safely

```
session.transaction(|session| {
    let value = session.new_string("committed")?;
    session.global_put("GemStoneRsCommitted", value)
})?;
```

The closure commits only when it returns `Ok`.

Recipe 5: Abort on Error

```
use gemstone_rs::Error;

let result: gemstone_rs::Result<()> = session.transaction(|session| {
    let value = session.new_string("will abort")?;
    session.global_put("GemStoneRsAborted", value)?;
    Err(Error::IllegalOop {
        operation: "intentional abort",
    })
});

assert!(result.is_err());
```

Recipe 6: Convert Rust Values to OOPs

```
use gemstone_rs::Value;

let seven = session.value_to_oop(&Value::SmallInt(7))?;
let yes = session.bool_oop(true);
let nothing = session.nil_oop();
```

Recipe 7: Fetch a String

```
let string_oop = session.new_string("hello")?;
let text = session.fetch_string(string_oop)?;
assert_eq!(text, "hello");
```

Recipe 8: Send a Message and Keep the Raw OOP

```
let seven = session.smallint_oop(7);
let printed_oop = session.perform_oop(seven, "printString", &[])?;
println!("{}", session.fetch_string(printed_oop)?);
```

Recipe 9: Send a Message and Marshal the Result

```
let seven = session.smallint_oop(7);
let value = session.perform(seven, "class", &[])?;
println!("{value:?}");
```

Recipe 10: Retain a Long-Lived OOP

```
let text = session.new_string("retained")?;
let handle = session.retain_oop(text)?;
println!("retained OOP {}", handle.oop().raw());
handle.release()?;
```

The handle releases automatically on drop if `release()` is not called.

Recipe 10A: Store a Rust Payload Under BridgeRoot

```
use gemstone_rs::{BridgeValue, Config, Session};
use std::collections::BTreeMap;

let mut session = Session::login(Config::from_env()?)?;
let labels = BTreeMap::from([("source".to_string(), "cookbook".to_string())]);
let payload = BridgeValue::dictionary([
    ("name".to_string(), BridgeValue::from("Tariq")),
    ("amount".to_string(), BridgeValue::from(100_i64)),
    ("currency".to_string(), BridgeValue::from("GBP")),
    ("labels".to_string(), BridgeValue::from(labels)),
]);

let mut bridge_root = session.bridge_root()?;
bridge_root.put("MyTestDict", payload)?;
bridge_root.commit()?;
```

The default root is `UserGlobals at: #GemStoneRsBridgeRoot`.

Recipe 10B: Round-Trip a Typed BridgeRoot Map Field

```
use gemstone_rs::{
    BridgeDictionary, BridgeFieldWrite, BridgeKeyType, BridgeMapped, BridgeValue,
Config, Session,
};
use std::collections::BTreeMap;

#[derive(Debug, Eq, PartialEq)]
struct BookingDraft {
    name: String,
    labels: BTreeMap<String, String>,
}

impl BridgeMapped for BookingDraft {
    fn to_bridge_value(&self) -> BridgeValue {
        BridgeValue::dictionary([
            ("name".to_string(), BridgeValue::from(self.name.clone())),
            ("labels".to_string(),
BridgeFieldWrite::to_bridge_field_value(&self.labels)),
        ])
    }

    fn from_bridge_dictionary(dictionary: &mut BridgeDictionary<'_>) ->
gemstone_rs::Result<Self> {
        Ok(Self {
            name: dictionary.at_string("name")?,
            labels: dictionary.at_map("labels")?,
        })
    }
}

let mut session = Session::login(Config::from_env()?)?;
let draft = BookingDraft {
    name: "Tariq".to_string(),
    labels: BTreeMap::from([("source".to_string(), "cookbook".to_string())]),
};

let mut bridge_root = session.bridge_root()?;
bridge_root.put_mapped("CookbookBookingDraft", &draft)?;
let loaded: BookingDraft = bridge_root.get_mapped("CookbookBookingDraft")?;
assert_eq!(loaded.labels["source"], "cookbook");

bridge_root.put_string("CookbookStatus", "ready")?;
bridge_root.put_vec("CookbookTags", &["priority".to_string(), "demo".to_string()])?;
bridge_root.put_map("CookbookLabels", &draft.labels)?;
assert_eq!(bridge_root.get_string("CookbookStatus")?, "ready");
let tags: Vec<String> = bridge_root.get_vec("CookbookTags")?;
assert_eq!(tags, vec!["priority".to_string(), "demo".to_string()]);
let labels: BTreeMap<String, String> = bridge_root.get_map("CookbookLabels")?;
assert_eq!(labels["source"], "cookbook");
```

```
bridge_root.put_map_with_key_type("CookbookLabelsSymbol", BridgeKeyType::Symbol,
&draft.labels)?;
let symbol_labels: BTreeMap<String, String> =
    bridge_root.get_map_with_key_type("CookbookLabelsSymbol",
BridgeKeyType::Symbol)?;
assert_eq!(symbol_labels["source"], "cookbook");
```

Use map fields for string-keyed metadata or lookup tables. Use nested `BridgeMapped` structs when the related value has a stable domain shape.

Recipe 11: Browse Dictionaries

```
use gemstone_rs::browser::Browser;

let mut browser = Browser::new(&mut session);
for dictionary in browser.dictionaries()? {
    println!("{dictionary}");
}
```

Recipe 12: Browse a Class

```
use gemstone_rs::browser::{Browser, ALL_PROTOCOLS};

let mut browser = Browser::new(&mut session);
let protocols = browser.protocols("Object", false, "")?;
let methods = browser.methods("Object", ALL_PROTOCOLS, false, "")?;
let source = browser.source("Object", "printString", false, "")?;
```

Pass an empty dictionary to resolve through the active user's symbol list.

Recipe 13: Diagnose a New Machine

```
gemstone-rs doctor
gemstone-rs doctor --live
gemstone-rs doctor --json
```

The first command checks environment and GCI loading, including whether `libgcirpc` came from explicit config, `GS_LIB_PATH`, `GS_LIB`, or `GEMSTONE/lib`, plus the exact path or directory searched. The second command logs in and verifies that `3 + 4` returns `7`. The JSON form is useful in CI or editor tooling.

Recipe 14: Inspect an OOP From the CLI

```
gemstone-rs inspect oop 20
```

This prints the raw OOP, class OOP, and `printString`.

Recipe 15: Inspect BridgeRoot From the CLI

```
gemstone-rs bridge root
gemstone-rs bridge keys
gemstone-rs bridge get BookingDraft --symbol
gemstone-rs bridge inspect BookingDraft --symbol
gemstone-rs bridge put-string WorkbenchDraft "hello from Rust"
gemstone-rs bridge put-symbol WorkbenchState ready
gemstone-rs bridge put-smallint WorkbenchCount 7
gemstone-rs bridge put-bool WorkbenchReady true
gemstone-rs bridge remove WorkbenchDraft
```

Use this when object-mapping examples or generated mappings write payloads under `GemStoneRsBridgeRoot`. `bridge keys` is the quickest way to see what is currently stored before inspecting a specific payload. The `put-*` shortcuts and `bridge remove` are useful live-write smoke tests because they commit one explicit BridgeRoot change at a time. Use generic `bridge put --type ...` when you want the value type to be data-driven in a script.

Recipe 16: Preview Codegen

```
gemstone-rs codegen preview examples/codegen/gemstone-rs.codegen
```

Use preview before writing generated files.

Recipe 17: Check Generated Code in CI

```
gemstone-rs codegen check examples/codegen/gemstone-rs.codegen
```

The repository runs the same check through `make verify`.

Recipe 18: Generate Wrappers

```
gemstone-rs codegen generate examples/codegen/gemstone-rs.codegen
```

Generated files are meant to be checked in.

Recipe 19: Discover a Config From a Live Stone

```
gemstone-rs codegen discover examples/codegen/discovered.codegen Object
```

Use this to bootstrap a config, then edit the generated mapping down to the API you actually want.

Recipe 19A: Run a Minimal Rust HTTP Health Service

```
cargo run -p gemstone-rs --example http_service -- --routes
cargo run -p gemstone-rs --example http_service -- --port 3000
```

From a second terminal:

```
curl -i http://127.0.0.1:3000/
curl -i http://127.0.0.1:3000/health/local
curl -i http://127.0.0.1:3000/health/gemstone
```

The example uses only the Rust standard library. It proves the service shape and GemStone health-check flow without pulling web framework dependencies into the core crate. `gemstone-py` is still ahead for batteries-included FastAPI, Litestar, and Django adapters; `gemstone-rs` now has a direct Rust service smoke path plus packaged Axum and Actix adapter crates:

```
cargo run --manifest-path examples/axum-service/Cargo.toml -- --routes
cargo run --manifest-path examples/actix-service/Cargo.toml -- --routes
```

The Axum and Actix services now start a bounded `SessionWorkerPool`, then use `gemstone-rs-axum` or `gemstone-rs-actix` to expose `/`, `/health/local`, and `/health/gemstone` without copying handler code.

For local development, the framework examples use the non-failing health-pool startup path. The server can start before credentials are configured: `/` and `/health/local` stay available, while `/health/gemstone` returns a `503` JSON error until the stone is reachable. The route smoke script checks that behavior without needing a live stone:

```
python3 scripts/framework_route_smoke.py
scripts/live_smoke.sh --dry-run
scripts/live_smoke.sh
```

The adapters also return `x-gemstone-rs-adapter`, `x-gemstone-rs-route`, `x-gemstone-rs-request-id`, `x-gemstone-rs-request-method`, and `x-gemstone-rs-request-path` headers, plus `x-gemstone-rs-request-lifecycle: received,handled` and `x-gemstone-rs-request-duration-us`. The smoke script asserts those headers so a proxy, load balancer, or test can distinguish the framework adapter, route, request, lifecycle, and handler duration that produced the response. It also asserts `x-gemstone-rs-example-middleware: axum` or `x-gemstone-rs-example-middleware: actix`, `x-gemstone-rs-service`, `x-gemstone-rs-service-version`, `cache-control: no-store`, and `x-content-type-options: nosniff` from the checked services. That proves the packaged routes still compose with normal framework middleware and gives the examples a small production-style cache/security header policy. In live mode the same script requires `/health/gemstone` to reach the stone and return `{"result":7}` for both adapters; the manual CI job runs that path with GemStone secrets.

Use `SessionWorker` when the application wants a reusable dedicated GemStone session lane instead of opening a new session inside each blocking route:

```
use gemstone_rs::{Config, SessionWorker, Value};

let worker = SessionWorker::start(Config::from_env()?)?;
assert_eq!(worker.eval("3 + 4")?, Value::SmallInt(7));
worker.shutdown()?;
# Ok::<(), gemstone_rs::Error>(())
```

Use `SessionWorkerPool` when the service should keep a fixed number of GemStone sessions warm and dispatch calls round-robin:

```
use gemstone_rs::{Config, SessionWorkerPool, Value};

let pool = SessionWorkerPool::start(Config::from_env()?, 2)?;
assert_eq!(pool.eval("3 + 4")?, Value::SmallInt(7));
pool.shutdown()?;
# Ok::<(), gemstone_rs::Error>(())
```

The same pool exposes awaitable calls for async runtimes:

```
use gemstone_rs::{Config, SessionWorkerPool, Value};

let pool = SessionWorkerPool::start(Config::from_env()?, 2)?;
assert_eq!(pool.eval_async("3 + 4").await?, Value::SmallInt(7));
pool.shutdown()?;
# Ok::<(), gemstone_rs::Error>(())
```

The installed CLI can also scaffold this shape:

```
gemstone-rs examples scaffold session_worker_pool ./gemstone-rs-worker-pool
```

The dependency-free `gemstone_rs::web` helpers now use the same async worker facade in Axum and Actix health handlers, so framework routes no longer need to wrap GemStone health checks in framework-specific blocking helpers.

Recipe 20: Check the Python Native Adapter Contract

Use this when you are preparing `gemstone-py-native` to wrap the Rust core:

```
gemstone-rs py-native capabilities
gemstone-rs py-native capabilities --json
gemstone-rs py-native check examples/py-native/gemstone-rs.py-native.json
gemstone-rs py-native smoke --dry-run
cargo run -p gemstone-rs --example python_native_adapter -- --dry-run
```

The CLI command prints the contract version, threading rule, supported operations, value kinds, error kinds, and OOP constants. The smoke command checks capabilities, OOP constants, value conversion, config error mapping, and structured error mapping without a live stone when `--dry-run` is passed. The live example logs in, evaluates `3 + 4`, performs `printString`, and round trips a `UserGlobals` string through `PyNativeSession`:

```
gemstone-rs py-native smoke
cargo run -p gemstone-rs --example python_native_adapter
```

The important point is architectural: Python should wrap `gemstone_rs::py_native`; it should not duplicate dynamic GCI loading or raw session calls.

Recipe 21: Run the Local Explorer

```
gemstone-rs-explorer --port 8787
```

Open:

```
http://127.0.0.1:8787/
```

The explorer starts read-only and loopback-only.

Recipe 22: Add a Local Explorer Auth Token

```
export GEMSTONE_RS_EXPLORER_TOKEN='replace-with-a-local-random-token'
gemstone-rs-explorer --port 8787 --auth-token-env GEMSTONE_RS_EXPLORER_TOKEN
curl -s 'http://127.0.0.1:8787/api/config?token=replace-with-a-local-random-token'
```

VS Code users can run `GemStone RS: Generate Explorer Auth Token`; the workbench stores the token in `gemstoneRs.explorerAuthToken`, launches the explorer with `--auth-token-env GEMSTONE_RS_EXPLORER_TOKEN`, and opens token-aware browser/webview URLs.

Recipe 23: Enable Explorer Eval Explicitly

```
gemstone-rs-explorer --port 8787 --allow-eval
curl -s 'http://127.0.0.1:8787/api/eval?source=3%20%2B%204'
```

Use this locally only.

Recipe 24: Use the VS Code Workbench With a Checkout

```
{
  "gemstoneRs.checkoutPath": "/path/to/gemstone-rs",
  "gemstoneRs.useCargo": true,
  "gemstoneRs.codegenConfig": "examples/codegen/gemstone-rs.codegen"
}
```

Then open the GemStone RS sidebar and use the Dictionaries, Codegen Config, and Explorer trees.

Recipe 25: Verify Published Artifacts

```
scripts/publish_verify.sh 0.2.2
```

The script checks crates.io versions, installs the CLI and explorer, runs `--help`, and checks the Marketplace version.

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